

Kathmandu Engineering College
Department of Electronics, Communication and Information



**Analysis, Spectrum Evaluation, and Demodulation
Variants of Amplitude Modulation Techniques**

Lab Report 4

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1 Objective

- To simulate Double Sideband Full Carrier (DSB-FC) modulation and evaluate envelope characteristics under varying modulation indices ($\mu = 0.5, 1.0, 1.5$).
- To compare the frequency spectra of standard Message, DSB-FC, and Double Sideband Suppressed Carrier (DSB-SC) systems using the Fast Fourier Transform (FFT).
- To generate Single Sideband Upper Sideband (SSB-USB) and Lower Sideband (SSB-LSB) signals via the Hilbert Transform phase-shift method.
- To implement Envelope and Product detection algorithms in MATLAB and evaluate their recovery performance for different AM variants.

2 Theory

2.1 Amplitude Modulation Variants

Amplitude Modulation shifts a low-frequency baseband message signal into a high-frequency passband channel by varying the peak value of a high-frequency carrier wave linearly with the message signal.

2.1.1 Double Sideband Full Carrier (DSB-FC)

Conventional AM (DSB-FC) transmits both information sidebands along with a prominent unmodulated carrier component. Given a message signal $m(t) = A_m \cos(2\pi f_m t)$ and carrier $c(t) = A_c \cos(2\pi f_c t)$, the modulated wave is:

$$s_{\text{DSB-FC}}(t) = A_c \left[1 + \mu \frac{m(t)}{A_m} \right] \cos(2\pi f_c t) \quad (1)$$

where $\mu = A_m/A_c$ is the modulation index. For distortion-free demodulation using simple envelope detectors, the condition $\mu \leq 1$ must be maintained. If $\mu > 1$ (over-modulation), the envelope crosses the zero-axis line, resulting in phase reversals that distort non-coherent recovery circuits.

2.1.2 Double Sideband Suppressed Carrier (DSB-SC)

To improve power efficiency, the unmodulated carrier component can be suppressed, leaving only the information-bearing sidebands. The modulation matches direct multiplication:

$$s_{\text{DSB-SC}}(t) = m(t) \cdot c(t) = A_m A_c \cos(2\pi f_m t) \cos(2\pi f_c t) \quad (2)$$

While DSB-SC channels direct all transmitted power into the sidebands, the lack of an explicit carrier reference means simple envelope detection is unusable. Coherent or synchronous demodulation is required.

2.1.3 Single Sideband (SSB-USB / SSB-LSB)

Since the upper and lower sidebands are symmetrical and carry identical message characteristics, transmission bandwidth can be reduced by 50% by removing one sideband entirely. Using the phase-shift approach, a -90° Hilbert transform $\hat{m}(t)$ is calculated. The Upper Sideband (USB) and Lower Sideband (LSB) are generated by:

$$s_{\text{SSB-USB}}(t) = m(t) \cos(2\pi f_c t) - \hat{m}(t) \sin(2\pi f_c t) \quad (3)$$

$$s_{\text{SSB-LSB}}(t) = m(t) \cos(2\pi f_c t) + \hat{m}(t) \sin(2\pi f_c t) \quad (4)$$

2.2 Demodulation Techniques

- **Envelope Detection:** Used for DSB-FC signals when $\mu \leq 1$. A half-wave rectifier path combined with an RC low-pass filter tracks the voltage peaks of the wave, extracting the message without needing an exact synchronous local carrier oscillator.
- **Product Detection:** Mandatory for suppressed carrier systems (DSB-SC and SSB). The incoming signal is multiplied by a locally generated carrier that matches the transmitter's frequency and phase. The mixed output is run through a low-pass filter to remove the high-frequency $2f_c$ component, isolating the baseband signal.

3 Source Code & Simulation Output Layouts

3.1 Setup and DSB-FC Modulation Index Variations

```

clc; clear all; close all;
fs = 10000;
t = 0:1/fs:0.1;
fc=1000; fm=100;
Am=1; Ac=1; mu = 0.5;

m=Am*cos(2*pi*fm*t);
c=Ac*cos(2*pi*fc*t);
N=length(t);
f_axis=(-N/2:N/2-1) * (fs/N);

% Part A: Modulation Indices
figure(1);
mu_vals=[0.5, 1.0, 1.5];
for i=1:3
    s=Ac*(1+mu_vals(i)* m/Am) .* c;
    subplot(3,1,i);
    plot(t,s,'b');
    title(sprintf('DSB-FC mu=%.1f - Buddha Shrestha', mu_vals(i)));
    xlabel('Time(s)'); ylabel('Amplitude')
;
    grid on;
end

```

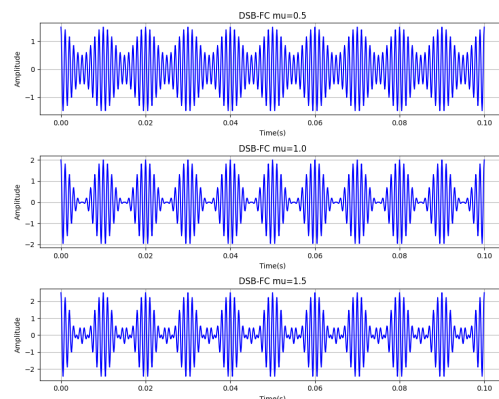


Figure 1: Time domain response under changing modulation indexes.

3.2 Frequency Spectra Evaluation

```

% Part B: Spectra Comparisons
am_fc = Ac*(1+mu*m/Am) .* c;
am_sc = m .* c;

figure(2);
subplot(3,1,1);
plot(f_axis,abs(fftshift(fft(m)/N)));
title('Message Spectrum - Buddha Shrestha');
xlim([-500 500]); grid on;

subplot(3,1,2);
plot(f_axis,abs(fftshift(fft(am_fc)/N)));
title('DSB--FC Spectrum - Buddha Shrestha');
xlim([-2000 2000]); grid on;

subplot(3,1,3);
plot(f_axis,abs(fftshift(fft(am_sc)/N)));
title('DSB--SC Spectrum - Buddha Shrestha');
xlim([-2000 2000]); grid on;

```

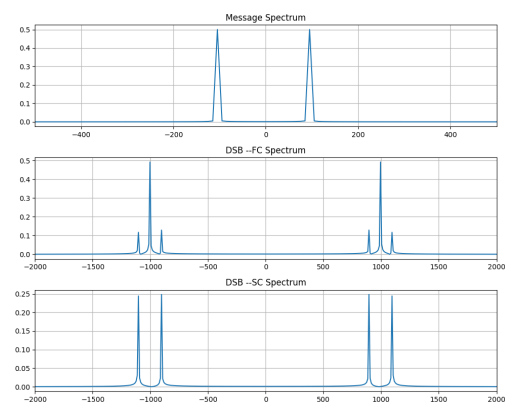


Figure 2: Fourier transform frequency spectra results.

3.3 Time Domain Representation of DSB-SC

```
% Part C: DSB-SC Signal Verification
figure(3);
subplot(3,1,1);
plot(t,m,'g');
title('Message - Buddha Shrestha');
grid on;

subplot(3,1,2);
plot(t,c,'k');
title('Carrier - Buddha Shrestha');
grid on;

subplot(3,1,3);
plot(t,am_sc,'b');
title('DSB-SC - Buddha Shrestha');
grid on;
```

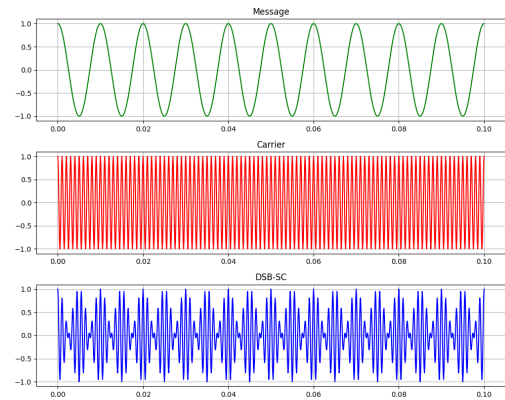


Figure 3: Time domain tracking for baseline DSB-SC structures.

3.4 Single Sideband (SSB) Generation and Spectra

```
% Part D: SSB Generation via Hilbert
Transform
m_hat = imag(hilbert(m));
ssb_usb=m.*cos(2*pi*fc*t) - m_hat.*sin(2*
pi*fc*t);
ssb_lsb=m.*cos(2*pi*fc*t) + m_hat.*sin(2*
pi*fc*t);

figure(4);
subplot(2,2,1); plot(t, ssb_usb,'b');
title('SSB_USB - Buddha Shrestha'); grid
on;

subplot(2,2,2); plot(t, ssb_lsb,'r');
title('SSB_LSB - Buddha Shrestha'); grid
on;

subplot(2,2,3); plot(f_axis, abs(fftshift(
fft(ssb_usb)/N)), 'b');
title('SSB_USB Spectrum - Buddha Shrestha'
); grid on;

subplot(2,2,4); plot(f_axis, abs(fftshift(
fft(ssb_lsb)/N)), 'r');
title('SSB_LSB Spectrum - Buddha Shrestha'
); grid on;
```

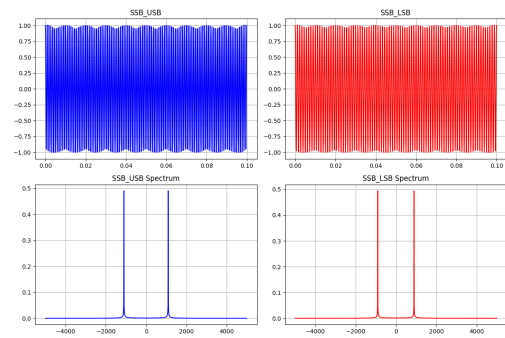


Figure 4: SSB analytical waveform outputs and asymmetric spectra.

3.5 Demodulation Loop and Signal Recovery

```
% Part E: Signal Recovery Loop
[b,a] = butter(5, 2*fm/fs*2);

env_det = filter(b,a,max(am_fc,0));
env_det = env_det - mean(env_det);
env_det=env_det/max(abs(env_det))*Am;

prod_sc = filter(b,a,am_sc.*c);
prod_sc=prod_sc/max(abs(prod_sc))*Am;

prod_ssb = filter(b,a,ssb_usb .* c);
prod_ssb = prod_ssb /max(abs(prod_ssb)) *
    Am;

figure(5);
subplot(3,2,1); plot(t,am_fc,'b');
title('DSB-FC Input - Buddha Shrestha');
grid on;
subplot(3,2,2); plot(t,m,'g', t, env_det,'
r--');
title('Envelope Detector'); legend('
original','Recovered'); grid on;

subplot(3,2,3); plot(t,am_sc,'b');
title('DSB-SC Input - Buddha Shrestha');
grid on;
subplot(3,2,4); plot(t,m,'g', t, prod_sc,'
r--');
title('Product Detector - DSB-SC'); legend
('original','Recovered'); grid on;

subplot(3,2,5); plot(t,ssb_usb,'b');
title('SSB-USB Input - Buddha Shrestha');
grid on;
subplot(3,2,6); plot(t,m,'g', t, prod_ssb,
'r--');
title('Product Detector - SSB'); legend('
original','Recovered'); grid on;
```

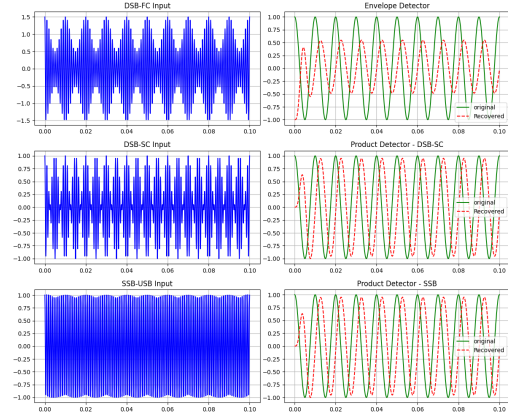


Figure 5: Comparison of recovered message signals against originals.

4 Discussion

The simulation plots illustrate the structural and performance trade-offs among different amplitude modulation schemes. Figure 1 demonstrates how the modulation index μ directly determines the envelope characteristics. For $\mu = 0.5$ and $\mu = 1.0$, the original message profile is preserved in the upper and lower boundaries of the carrier signal. When over-modulated at $\mu = 1.5$, the envelope crosses zero, causing 180° phase reversals. This explains why standard diode envelope detectors fail on over-modulated inputs, introducing non-linear harmonic clipping.

The frequency-domain analysis highlighted differences in bandwidth and power efficiency. The DSB-FC spectrum exhibits a prominent unmodulated carrier spike at 1000 Hz, which consumes up to two-thirds of total transmission power without carrying data. In the DSB-SC spectrum, this central spike is removed, focusing all power into the sidebands (900 Hz and 1100 Hz). The SSB plots show that the Hilbert transform method effectively suppresses one sideband, reducing the required transmission bandwidth by half (100 Hz instead of 200 Hz) while keeping the message information intact.

Finally, the demodulation loops showed that receiver complexity depends on the chosen modulation scheme. Simple envelope detection works well for under-modulated DSB-FC signals, using basic rectification and low-pass filtering. Suppressed carrier formats (DSB-SC and SSB), however, require a synchronous product detector. Even minor phase offsets in the local oscillator can cause phase attenuation or complete signal cancellation, confirming that suppressed carrier architectures improve transmission efficiency at the expense of requiring more complex receiver hardware.

5 Conclusion

This lab experiment successfully analyzed the generation, spectral distribution, and demodulation of various amplitude modulation techniques (DSB-FC, DSB-SC, and SSB) using MATLAB. The results verified that conventional DSB-FC allows for simple, low-cost receiver design via envelope detection, provided the modulation index satisfies $\mu \leq 1$. Suppressed-carrier techniques (DSB-SC and SSB) eliminate carrier power wastage and reduce transmission bandwidth, but require complex, synchronous product detectors. The experiment successfully demonstrated the fundamental engineering trade-offs between transmitter power efficiency, spectral bandwidth, and receiver circuit complexity.